# De La Salle Holy Cross College

**NOVEMBER EXAMINATIONS**

2023

Grade 10

# MATHEMATICS

Paper 2

| | | | |
|---|---|---|---|
| **Examiner:** | A. Seewald | **Time:** | 2 hours |
| **Moderators:** | A. Bux, K. Ruperto & J. Blockley | **Marks:** | 100 |

| Name: Final Draft | | | | |
|---|---|---|---|---|
| Teacher: | A. Bux | A. Seewald | K. Ruperto | J. Blockley |

**PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY**

1. This question paper consists of:
   - 16 pages that includes 1 page for extra calculations or rough work,
   - 11 questions and a separate information sheet on page 2.
2. Please check that your question paper is complete.
3. Answer the questions in the space provided on this question paper.
4. A non-graphical, non-programmable calculator may be used. Ensure that it is set in DEGREE mode.
5. ALL necessary calculations must be shown clearly. Answers only will not necessarily be awarded full marks.
6. Diagrams and graphs are not necessarily drawn to scale.
7. Round off final answers to ONE decimal place, unless stated otherwise.
8. It is in your own interest to write legibly and to present your work neatly.
9. One blank page (page 16) is included in the paper. If you run out of space for a question, then use this page. Clearly indicate the question number of your answer should you use this extra space.

| | Q 1 | Q 2 | Q 3 | Q 4 | Q 5 | Q 6 | Q 7 | Q 8 | Q 9 | Q 10 | Q 11 | Total |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mark | JB | AB | AB | AB | JB | JB | JB | AS | AS | AS | AS | |
| Total | 9 | 14 | 17 | 4 | 8 | 10 | 4 | 5 | 11 | 11 | 7 | 100 |

# Mathematics Formula Sheet

## Grade 10 2023

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M = \left(\frac{x_1 + x_2}{2}; \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$\tan\theta = m$$

$$\bar{x} = \frac{\Sigma f.x_i}{n}$$

## SECTION A

### QUESTION 1 [ 9 Marks ]

The number of trips that **32** taxi drivers undertake in one week are as follows:

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 59 | 74 | 40 | 58 | 35 | 54 | 42 | 55 | 48 | 61 | 80 | 73 |
| 36 | 82 | 65 | 68 | 39 | 54 | 41 | 66 | 70 | 45 | 50 | 59 |
| 81 | 44 | 63 | 32 | 47 | 62 | 72 | 43 | | | | |

(a) Complete the following table: (4)

| Interval | Tally | Frequency | Class Midpoint | $f \times x$ |
|---|---|---|---|---|
| $30 \leq x < 40$ | IIII | 4 | 35 | 140 |
| $40 \leq x < 50$ | ~~IIII~~ III | 8 | 45 | 360 |
| $50 \leq x < 60$ | ~~IIII~~ II | 7 | 55 | 385 |
| $60 \leq x < 70$ | ~~IIII~~ I | 6 | 65 | 390 |
| $70 \leq x < 80$ | IIII | 4 | 75 | 300 |
| $80 \leq x < 90$ | III | 3 | 85 | 255 |

✓ ✓ ✓ ✓

(b) Calculate the approximate mean number of trips for the grouped data. (3)

$$\text{mean} = \frac{1830}{32} = 57,2$$ ✓ ✓ ✓ ✓ ✓ ca

(c) Determine the modal class. (1)

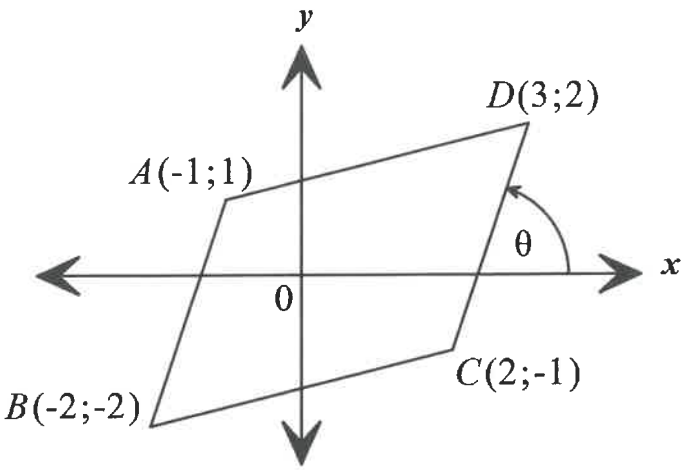
$40 \leq x < 50$ ✓

(d) In which interval does the median lie? (1)

$50 \leq x < 60$ ✓

**QUESTION 2** **[ 14 Marks ]**

Quadrilateral ABCD has vertices A(-1; 1), B(-2; -2), C(2; -1) and D(3; 2).




(a) Determine the lengths of AB and CD. (Leave your answers in simplest surd form) (3)

$d_{AB} = \sqrt{(-2+1)^2 + (-2-1)^2}$ ✓m

$= \sqrt{10}$ units ✓

$d_{CD} = \sqrt{(3-2)^2 + (2+1)^2}$

$= \sqrt{10}$ units ✓

(b) Determine the gradients of AB and CD. (3)

$m_{AB} = \frac{-2-1}{-2+1} = \frac{-3}{-1} = 3$ ✓m ✓

$m_{CD} = \frac{2+1}{3-2} = \frac{3}{1} = 3$ ✓

(c) What can you deduce about lines AB and CD from your answer in (b)? (1)

$\therefore m_{AB} = m_{CD} = 3$

$\therefore AB \parallel CD$ ✓

(d) What type of quadrilateral is ABCD? Give a reason for your answer. (2)

Parallelogram. One pair of sides equal and parallel.

(e) Determine the equation of line perpendicular to AB that intercepts with point C. (3)

$m_{AB} = 3$

$\therefore m_{AC} = -\frac{1}{3}$

$\therefore y = -\frac{1}{3}x + c$

sub $(2;-1)$ : $-1 = -\frac{1}{3}(2) + c$ sub

$-1 = -\frac{2}{3} + c$

$-\frac{1}{3} = c$

$\therefore y = -\frac{1}{3}x - \frac{1}{3}$

(f) Calculate the value of $\theta$. (2)

$\therefore \tan\theta = m_{CD}$

$\tan\theta = 3$

$\therefore \theta = 71,57°$

**QUESTION 3** **[ 17 Marks ]**

(a) Use a calculator to find the values of the following, **give your answer correct to 3 decimal places**:

(1) $2 \sin 105°$ (1)

$= 1,932$

(2) $\cot 61,4°$ (2)

$= \frac{1}{\tan(61,4°)}$

$= 0,545$

(b) Determine the value of $\theta$.

(1) $\tan\theta = 2{,}461$ (1)

$\theta = 67{,}9°$ ✓

(2) $2\sin 3\theta = \sqrt{3}$ (3)

$\sin 3\theta = \frac{\sqrt{3}}{2}$ ✓

$3\theta = 60°$ ✓

$\therefore \theta = 20°$ ✓

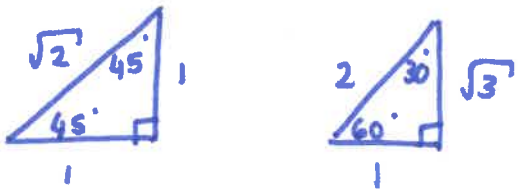
(c) Determine the value of the following **without the use of a calculator**: (4)

$$\frac{\sin 60°.\tan 30°}{\sec 45°}$$

$= \frac{\frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{3}}}{\sqrt{2}}$ ✓ ✓ ✓

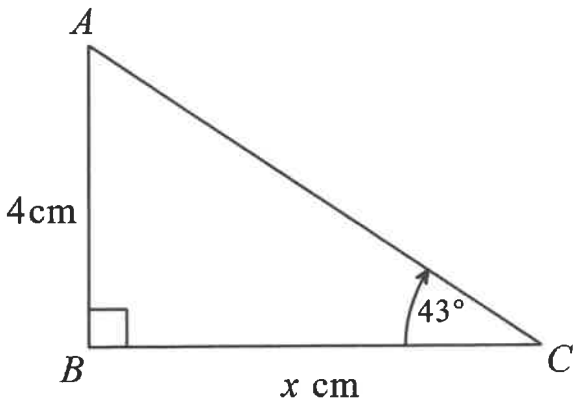
$= \frac{1}{2} \div \frac{\sqrt{2}}{1}$

$= \frac{1}{2\sqrt{2}}$ ✓




(d) Solve for $x$ in the following triangles:

(1) (3)

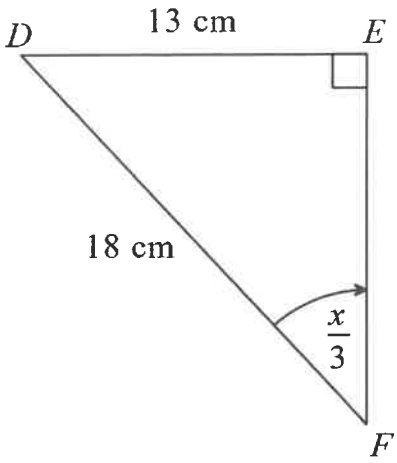



$\tan 43° = \frac{4}{x}$ ✓

$\therefore x = \frac{4}{\tan 43°}$ ✓

$\therefore x = 4{,}3\text{cm}$ ✓

(2)




(3)

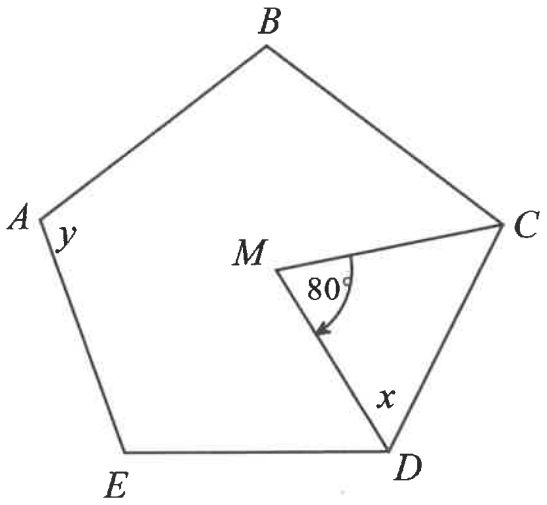
$\sin\left(\frac{x}{3}\right) = \frac{13}{18}$ ✓

$\therefore \frac{x}{3} = 46,2$ ✓

$\therefore x = 138,7°$ ✓

**QUESTION 4** **[ 4 Marks ]**

ABCDE is a regular pentagon. M is equidistant from C and D.




Determine the size of:

(a) $x$ (2)

$x = \frac{180° - 80°}{2}$ ∠'s opp = sides ✓S+R

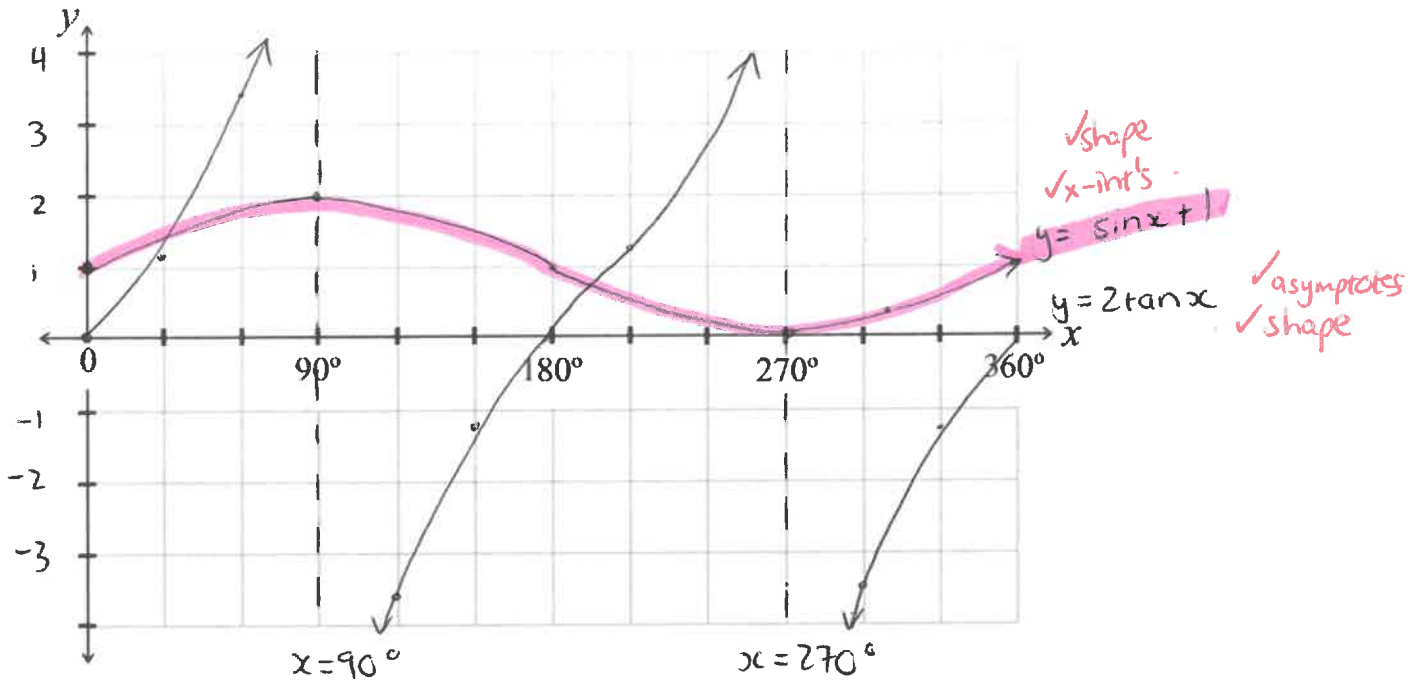
$x = 50°$ ✓

(b) $y$ (2)

$y = \frac{3(180°)}{5} = 108°$ ✓✓

**QUESTION 5** **[ 8 Marks ]**

(a) Sketch the graphs of $f(x) = 2\tan x$ and $g(x) = \sin x + 1$ for $x \in [0°; 360°]$.
Clearly indicate all intercepts with the axes and any asymptotes. (4)




(b) What is the period of $f(x) = 2\tan x$? (1)

180° ✓

(c) Determine the range of $g$. (1)

$y \in [0°; 2]$ ✓

(d) $h(x)$ is the reflection of $g(x)$ about the $x$-axis and a shift of 1 unit up.
Write down the equation of $h(x)$. (1)

$y = -\sin x + 2$ ✓

(e) Determine the $x$ values where $f(x) > 0$. (1)

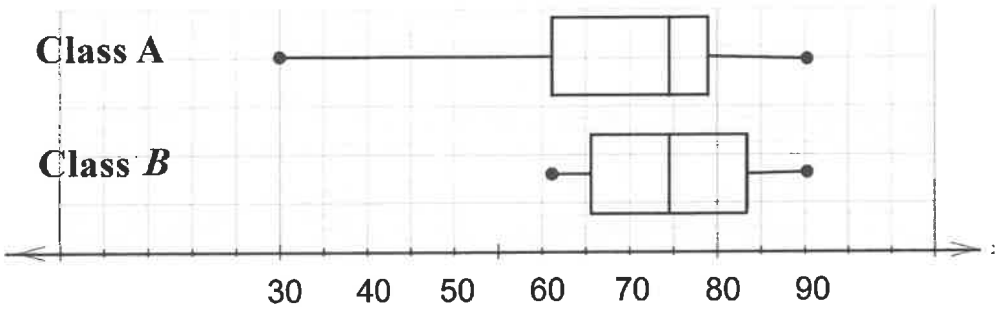
$x \in (0; 90°) \cup (180°; 270°)$ ✓both

**TOTAL SECTION A: 52**

## SECTION B

## QUESTION 6 [ 10 Marks ]

(a) The box and whisker plots below summarise the final test scores for two of Mr Smith's Mathematics classes from the same grade.




(1) Describe the features in the scores that are the same for both classes. (2)

median & maximum ✓ ✓

(2) Calculate the interquartile range for Class B. (1)

IQR = 84 − 66 = 18 ✓

(or 17 accepted)

(3) Mr Smith considers the median of each class and reports that there is no significant difference in the performance between them.
Is Mr Smith's conclusion valid? Support your answer with **reasons**. (2)

No. ✓

Class A is a lot weaker than Class B.

Class A has a very low minimum mark compared to class B.

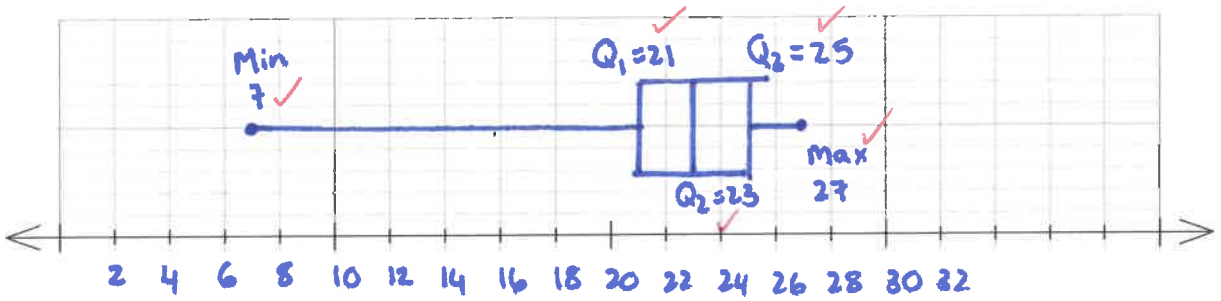
IQR of class A < IQR class B

Class A is skew to left, ie. lower marks than class B

Any valid reason. ✓

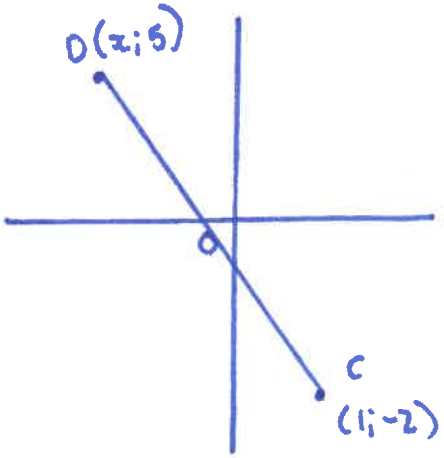
(b) Draw a box and whisker diagram to represent a set of data if the: (5)

- Median is 23
- Lower Quartile is 21
- IQR is 4
- Top 25% of the data lies between 25 and 27
- Range is 20




**QUESTION 7** **[ 4 Marks ]**

C is the point (1; -2). The point D lies in the second quadrant and has co-ordinates ($x$; 5).
If the length of CD is $\sqrt{53}$ units, calculate the value of $x$.




$CD^2 = (x-1)^2 + (5+2)^2$ ✓m

$(\sqrt{53})^2 = (x-1)(x-1) + 7^2$

$53 = x^2 - 2x + 1 + 49$

$0 = x^2 - 2x - 3$ ✓

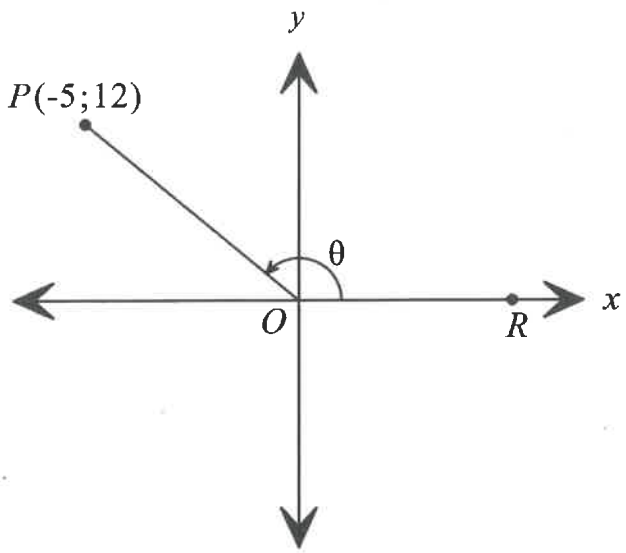
$0 = (x-3)(x+1)$ ✓ factors

$\therefore x = 3$ or $x = -1$ ✓a

N/A

**QUESTION 8** **[ 5 Marks ]**

In the diagram, $P(-5;12)$ is a point in the Cartesian plane and $R\hat{O}P = \theta$.




Determine the value of:

(a) $\cos\theta$ (2)

$= -\frac{5}{13}$ ✓✓

$r = 13$ Pythag

(b) $\text{cosec}^2\theta + 1$ (3)

$\frac{1}{\left(\frac{12}{13}\right)^2} + 1$ ✓

$= \frac{169}{144} + 1$ ✓

$= \frac{313}{144}$ ✓

$= 2,17$

$x = -5$

$y = 12$

$r = 13$

$\text{cosec}\theta = \frac{1}{\sin\theta}$

$= \frac{1}{\frac{y}{r}}$

$= 1 \div \frac{12}{13}$

$= 1 \times \frac{13}{12}$

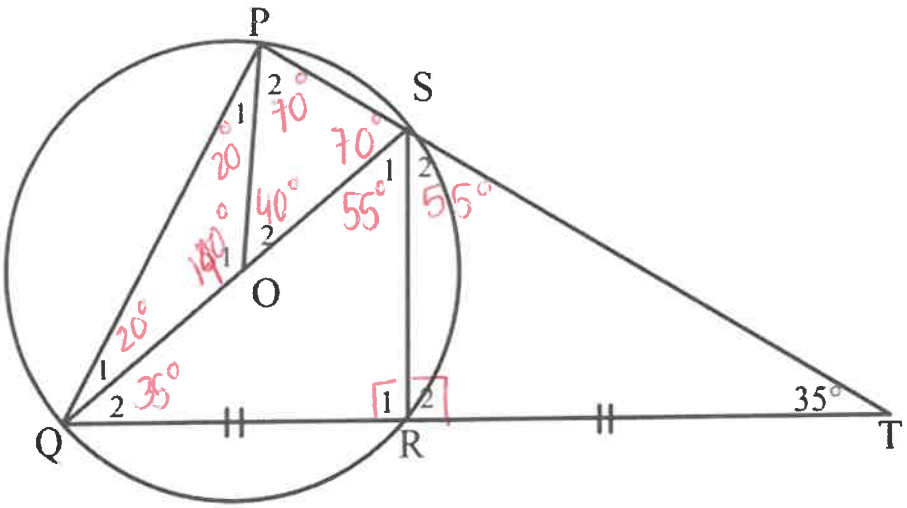
$= \left(\frac{13}{12}\right)^2 = \frac{169}{144}$

$= 1,17$

**QUESTION 9** **[ 11 Marks ]**

QOS is a diameter of the circle with centre O.

- QR = RT
- $\hat{T} = 35°$




(a) Calculate, with reasons, the size of:

(1) $\hat{R}_2$ (2)

$\hat{R}_1 = 90°$ L's in semi-circle ✓str

$\therefore \hat{R}_2 = 90°$ L's on a str. line. ✓str

(2) $\hat{S}_2$ (2)

$\hat{S}_2 = 180° - 35° - 90°$ L's in a Δ ✓ste

$\hat{S}_2 = 55°$ ✓a

(b) Prove $\Delta SQR \equiv \Delta STR$ (4)

In $\Delta SQR$ and $\Delta STR$

$SR$ common ✓

$\hat{R}_1 = \hat{R}_2$ proven ✓

$QR = RT$ given ✓

$\therefore \Delta SQR \equiv \Delta STR$ (SAS) ✓

(c) Calculate, with reasons, the size of:

(1) $\hat{Q}_1$ (2)

$\hat{Q}_2 = \hat{T} = 35°$ ($\Delta SQR \equiv \Delta STR$) ✓S+R

$\therefore \hat{Q}_1 + \hat{Q}_2 = \hat{S}_2$ ext. ∠ of quad ✓S+R.

$\therefore \hat{Q}_2 = 20°$

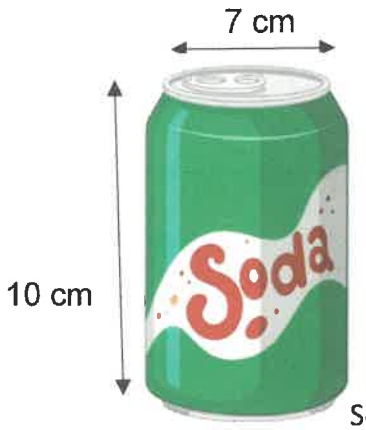
$\hat{Q}_1 = 20°$.

(2) $\hat{P}_1$ $= 20°$ isos. Δ; radii. (1)

✓S+R.

**QUESTION 10** **[ 11 Marks ]**

A can of cooldrink has measurements as shown in the figure.




Source: Freepik.com

SA of closed cylinder = $2\pi r^2 + 2\pi rh$

Volume of cylinder = $\pi r^2 h$

Determine:

(a) The total exterior surface area of the can. (4)

$SA = 2\pi r^2 + 2\pi rh$ $r = 3{,}5\text{cm}$

$= 2\pi(3{,}5)^2 + 2\pi(3{,}5)(10)$

$= 296{,}9\ \text{cm}^2$

(b) The volume of the can. (2)

$V = \pi r^2 \times h$

$= \pi(3{,}5)^2 \times 10$

$= 385\ \text{cm}^3$ $384{,}8\ \text{cm}^3$

(c) How many cans can be packed in a box with a length of 28 cm, a width of 21 cm and a height of 20cm? (3)

$4 \times 3 \times 2$

$= 24$ tins

(d) If the radius of the can is halved, what will the volume of the smaller can be? (2)

$V \times (\text{scale factor})^2$

$= 385 \times \left(\frac{1}{2}\right)^2$

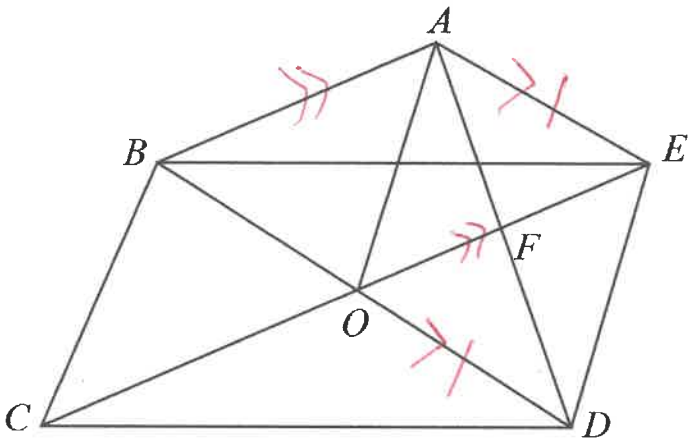
$= 96{,}2\ \text{cm}^3$

or $V_{new} = \pi(1{,}75)^2 \times 10$

$= 96{,}2\text{cm}^3$

**QUESTION 11** **[ 7 Marks ]**

In the diagram, BCDE and AODE are parallelograms.




(a) Prove that OF ∥ BA. (3)

O is midpt of BD  Diagonals of parm bisect ✓STR

F is midpt of AD  Diagonals of parm bisect ✓STR

∴ OF ∥ AB  Midpt. Thm. ✓R

(b) Prove that ABOE is a parallelogram. (4)

AE ∥ OD  opp. sides of parm ✓STR

∴ AE ∥ BO ✓∥

and CF ∥ AB  proven ✓STR

∴ OE ∥ AB ✓

∴ ABOE is a parm  both pairs of opp. sides parallel ✓R

**TOTAL SECTION B: 50**